ARITRA MUKHOPADHYAY

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Summary

B.Tech CSE (AI/ML) undergraduate with hands-on experience in machine learning, deep learning, computer vision, and Generative AI. Developed RAG applications, intelligent web systems, and ML-powered solutions using Python, PyTorch, TensorFlow, React, and modern AI technologies. Interested in applied AI research and scalable ML systems.

Education

The Neotia University 2023 – 2027
B.Tech in Computer Science and Engineering (AIML) **CGPA: 9.15**

Vivekananda Vidyapith (H.S) 2022 – 2023
XII (WBCHSE) **95.28%**

Experience

AI/ML Research Intern Jun 2026 – Jul 2026
National Institute of Technology (NIT), Durgapur, India

- Investigated 7 EfficientNet-B0 variants on CIFAR-100 using TensorFlow, replacing standard convolutions with quadratic convolution formulations.
- Evaluated RBF, Polynomial, Swish, Leaky ReLU, and ELU activations; achieved **84.29% accuracy**, a **5.57 percentage-point improvement** over the 78.72% baseline, and analyzed gradient stability.

Generative AI Intern [Jun 2026 – Jul 2026]
GENBYTE AI, Kerala, India

- Built a RAG-based Tourism Assistant using Groq (Llama 3.3 70B), Streamlit, and Supabase pgvector; developed semantic retrieval with 384-dimensional MiniLM embeddings and cosine similarity.
- Implemented PDF, TXT, and Markdown ingestion, source-level citations, security guardrails, local vector fallback, and automated evaluation across 5 tourism benchmark categories.

Advanced Linux Intern Jun 2025 – Jul 2025
Webel Pvt. Ltd., India

- Practiced Linux system administration, file system management, user/group permissions, process control, command-line tools, and shell scripting.
- Configured Linux environments and explored system architecture, resource management, task automation, and troubleshooting.

Projects

MANTRA – AI-Powered Task Execution and Verification System Live Demo

- **Tech Stack:** React 18, Vite, JavaScript, Groq API, NVIDIA API, LLMs, Multimodal AI, Computer Vision.
- Developed an AI task guidance platform that converts natural-language instructions into actionable checklists, with an integrated AI assistant for contextual guidance.
- Integrated Llama 3.3 70B for task generation and Llama 4 Scout for multimodal image analysis; implemented MediaDevices camera access for workspace scanning, visual step verification, and final quality assessment.

NearNest – AI-Powered Family Location Tracking System GitHub

- **Tech Stack:** Kotlin, Android, Jetpack Compose, Gemini API, FusedLocationProviderClient, Room SQLite, Retrofit.
- Developed a native Android family safety application with real-time GPS tracking, geofenced safe zones, location history, and movement monitoring.
- Integrated Gemini for contextual analysis of verified JSON location trajectories with privacy and anti-hallucination guardrails; implemented offline-first Room storage, batch synchronization, and 30-day retention.
- Validated sub-2-second synchronization performance across 8 test scenarios.

Stress Shield – AI-Based Occupational Stress Monitoring System Live Demo

- **Tech Stack:** Flutter, React, TypeScript, Python, FastAPI, Node.js, PostgreSQL, Docker, BLE, Groq API.
- Developed a cross-platform biometric monitoring system integrating smartwatch telemetry through BLE GATT, Android Health Connect, and Apple HealthKit.
- Built a physiological monitoring pipeline for Heart Rate, HRV/RMSSD, GSR, strain, and duty-cycle indicators to support occupational stress assessment.
- Developed a FastAPI-based ML service and Node.js backend with PostgreSQL; integrated Groq-powered AI risk triage for recovery recommendations and supervisor alerts.

Skills

Programming: Python, Java, C, SQL, R, JavaScript **Core CS:** OOP, DBMS, OS, DSA, Computer Networks
AI/ML: TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, NLTK
Generative AI: RAG, LLMs, Semantic Search, Vector Databases, Prompt Engineering, Groq API
Frontend: React.js, Next.js, Tailwind CSS, HTML, CSS **Backend:** Node.js, FastAPI, Flask, Express.js, Streamlit
Databases: PostgreSQL, Supabase, MySQL **Tools:** Docker, Git, GitHub, Linux, AWS, Vercel, Render